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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    confusion_matrix,
    classification_report,
    accuracy_score,
    precision_score,
    recall_score,
    roc_auc_score,
    roc_curve
)

# -----
# Load Dataset
# -----
df = pd.read_csv("data.csv")

# -----
# Data Preprocessing
# -----

# Remove unnecessary columns
df.drop(["id", "Unnamed: 32"], axis=1, inplace=True)

# Convert target column into numeric
# M = 1 (Malignant)
# B = 0 (Benign)
df["diagnosis"] = df["diagnosis"].map({"M":1, "B":0})

# Features and Target
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

# -----
# Train-Test Split
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# -----
# Feature Scaling
# -----
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# -----
# Logistic Regression Model
# -----
model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

# -----
# Predictions
# -----
y_pred = model.predict(X_test)

# Probability predictions
y_prob = model.predict_proba(X_test)[:,:1]

# -----
# Evaluation Metrics
# -----
print("Accuracy :", accuracy_score(y_test, y_pred))

print("\nPrecision :", precision_score(y_test, y_pred))

print("\nRecall :", recall_score(y_test, y_pred))

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print("\nROC-AUC Score :", roc_auc_score(y_test, y_prob))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# -----
# Confusion Matrix
# -----
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=["Benign", "Malignant"],
    yticklabels=["Benign", "Malignant"]
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

# -----
# ROC Curve
# -----
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

plt.figure(figsize=(8,5))
plt.plot(fpr, tpr, label="ROC Curve")
plt.plot([0,1], [0,1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

# -----
# Correlation Heatmap
# -----
plt.figure(figsize=(15,10))
sns.heatmap(df.corr(), cmap="coolwarm")
plt.title("Feature Correlation Heatmap")
plt.show()

# -----
# Basic Insights
# -----
print("\nInsights:")
print("1. Logistic Regression successfully classifies tumors.")
print("2. ROC-AUC close to 1 indicates excellent performance.")
print("3. High precision means fewer false positives.")
print("4. High recall means fewer false negatives.")
print("5. Standardization improves model performance.")

```

